INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
(School of University of Engineering and Management, Kolkata)

End Semester Examination, November - December, 2025

Degree: **B.Tech** Semester: **First**

Stream: **BT / CSE(AI)/ CSE(AI & ML)/ CSE(Cybersecurity)/ CSE (IOT, CS, BT)/ CSE(Networks)/ CSBS/ CSE/ ECE/ EEE/ EE/ IT/ ME**

Paper Code: **BSCM103B**

Paper Name: **Mathematics and Basic Statistics**

Full Marks: **100** Duration: **3 Hours**

Part - A
Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1. Find whether the set {(1,0,0), (0,1,0), (8,- 1, 0)} is linearly independent in the vector space R^3. **2**

2. Find whether the set $S = \{(x, -x) \epsilon R^2\}$ is a subspace of R^2. **2**

3. The set of vectors S = {(1,0,0), (1,2,3), (10,10,12), (5,6,7)} is a basis of R^3- comment. **2**

4. Find a linear transformation T : $V_3 \rightarrow V_4$, whose image is generated by the vectors (1, 2, 0, -4) and (2, 0, -1, -3). **2**

5. Check whether the following transformation is linear transformation or not. **2**
$T : R^3 \rightarrow R^3$ defined by T (x, y, z) = (x + y, y + z, x + y)

6. In R^3 let $\alpha = (x_1, x_2, x_3)$, $\beta = (y_1, y_2, y_3)$ then find whether (α, β) is defined as (α, β) is an inner product on R_3 if (α, β) is defined as $(\alpha, \beta) = x_1y_1 + (x_2 + y_3)(y_2 + y_3)$. **2**

7. Find the differential equations of all parabolas with x-axis as the axis and (a, 0) as focus. **2**

8. What is the solution of $\frac{d^3y}{dt^2} - x = 0$? **2**

9. The C.F. of $y'' - 2y' + y = xe^x \sin x$ is = ? **2**

10. What is the solution of the differential equation $y'' + y = 0$? Satisfying the conditions $y(0) = 1$ and $y(\pi/2) = 2$. **2**

11. Determine k so that the vectors (1,3,1), (2,k,0) and (0,4,1) are linearly dependent in R^3. **2**

12. The Wronskian of x and e^x is = ? **2**

13. Let T : V → W be a linear transformation. **2**
Prove that $T(\alpha) = T(\beta) \Rightarrow \alpha = \beta$ if Ker (T) = {θ}.

14. Find the mean, median and mode for the following: **2**

Mid Value :	15	20	25	30	35	40	45	50	55
Frequency :	2	22	19	14	3	4	6	1	1

15. Form the differential equation from the following equation **2**
$x = a \sin(at + b)$

Part - B
Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Let V be vector space over a field F with basis $\{\alpha_1, \alpha_2, \alpha_3\}$. T : V → V in a linear transformation such that $T(\alpha_1) = \alpha_2$, $T(\alpha_2) = \alpha_3$, $T(\alpha_3) = \theta$. Prove that T^3 is a zero transformation. **5**

17. $T : R^3 \rightarrow R^3$ defined by **5**
$T(x_1, x_2, x_3) = (x_1 + 2x_2 + 3x_3, 2x_1 + 3x_2 + x_3, 3x_1 + x_2 + 2x_3)$.
Find matrix of T relative to the basis. {(-1,1,1), (1,-1,1), (1,1,-1)} of R^3

18. Solve $\frac{dy}{dx} = e^{3x-2y} + x^2 e^{-2y}$. **5**

19. Solve $(D^2 - 4D + 4)y = x^{2x} + \cos 2x + 4$. **5**

20. Find the correlation coefficient between x and y given data: **5**

x :	78	89	97	69	59	79	68	57
y :	125	137	156	112	107	138	123	108

21. Use Gram-Schmidt process to obtain an orthogonal basis from the following basis set of the relevant inner product space. Then convert it to orthonormal basis (with standard inner product). {(1,1), (-1,2)} of R^2 **5**

22. Solve $\frac{dz}{dx} + \left(\frac{z}{x}\right) \log z = \frac{z}{x} (\log z)^2$ **5**

23. Solve $\left(y^2 e^{xy^2} + 4x^3\right)dx + \left(2xy\, e^{xy^2} - 3y^2\right)dy = 0$. **5**

24. For the distribution: **5**

x:	5	6	7	8	9	10	11	12	13	14	15
y:	18	15	34	47	68	90	80	62	35	27	11

Find the mean, median, and lower and upper quartiles, variance and the standard deviation.

Part - C

Attempt any 5 questions out of 8 questions

Each question carries 10 Marks (10 × 5)

25. Show that if $\lambda \neq -5$, the system of equations $3x - y + 4z = 3,\ x + 2y - 3z = -2,\ 6x + 5y + \lambda z = -3$, Have a unique solution. If $\lambda = -5$, show that the equations are consistent, Determine the solutions in each case. **10**

26. Find the characteristic equation of the matrix $A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$. Show that the equation is satisfied by A and hence obtain the inverse of the given matrix. **10**

27. Solve $xy\,(1 + xy^2)\,\frac{dy}{dx} = 1$. **10**

28. Find the complete solution of $y'' - 2y' + 2y = x + e^x \cos x$. **10**

29. Prove that the set of all polynomials of degree less or equal to n is a vector space with respect to usual addition between two polynomials (as defined in the section 'polynomial Ring') and usual multiplication between a number and a polynomial. **10**

30. Show that {(1,2,1), (0,1,0), (0,0,1)} is a basis of R^3 **10**

31. Find the rank correlation for the following data : **10**

x:	56	42	72	36	63	47	55	49	38	42	68	60
y:	147	125	160	118	149	128	150	145	115	140	152	155

32. The heights and weights of the 10 army men are given below, In which characteristics are they more variable? **10**

Height in cm.	170	172	168	177	179	171	173	178	173	179
Weight in kg.	75	74	75	76	77	73	76	75	74	75

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